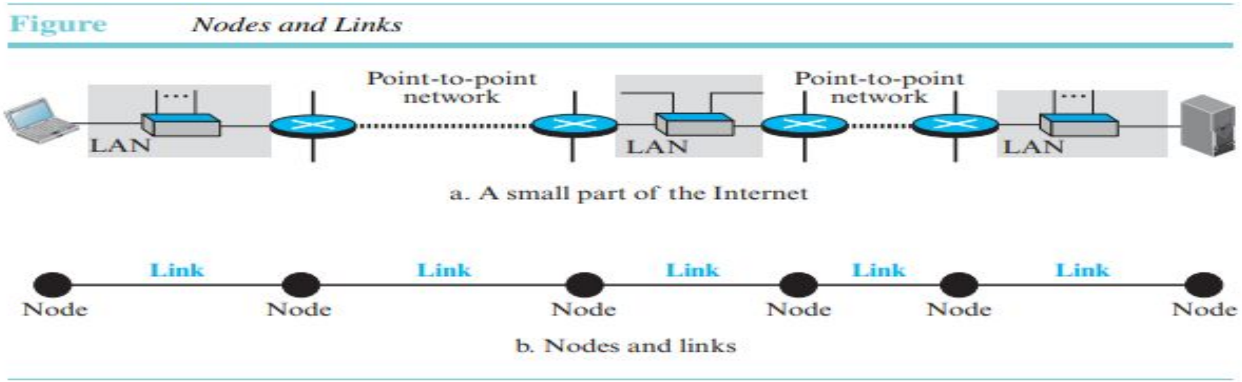
COMPUTER NETWORKS

IV MODULE - DATA LINK LAYER

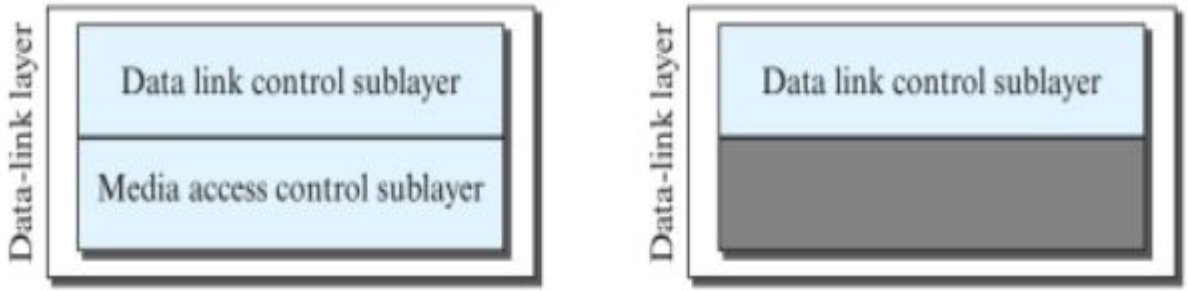
Data link layer

- Enables node to node / hop-to-hop communication



- Two Types of Links: point to point, broadcast links
- 2 sublayers: Logical link control(LLC), Media Access Control(MAC)

Figure 5.3 *Dividing the data-link layer into two sublayers*

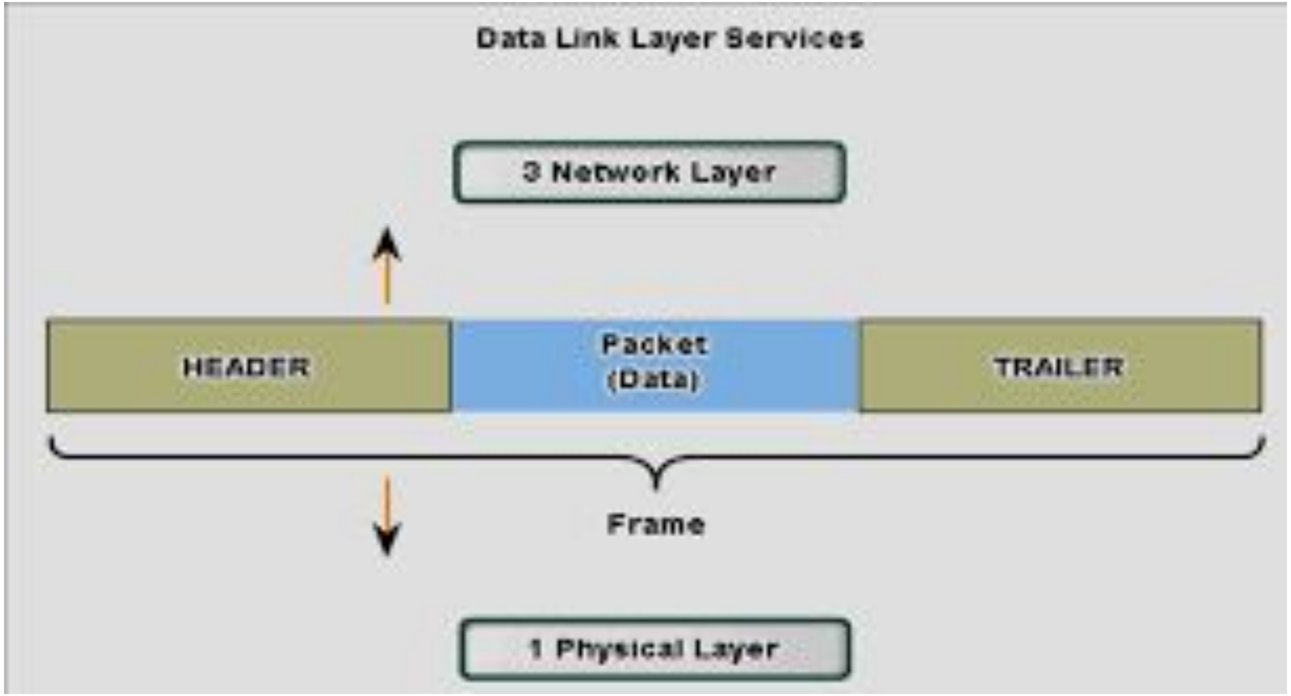


a. Data-link layer of a broadcast link

b. Data-link layer of a point-to-point link

LLC: it controls the framing, flow control, error control and error detection and correction functions of the data link layer.

MAC: It controls the transmission of data packets via shared channels.


Data Link Layer Services
3 Network Layer
HEADER
Packet
(Data)
TRAILER
Frame
1 Physical Layer

Services provided by Link layer

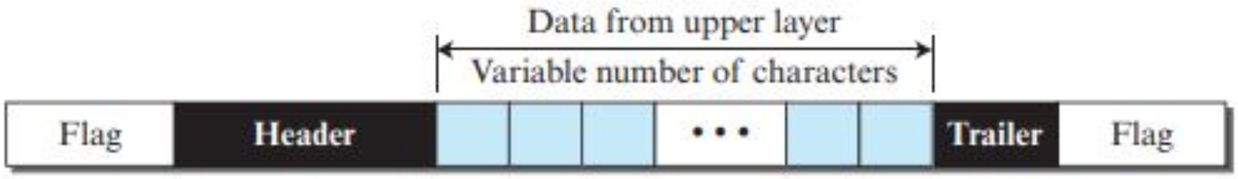
Framing:

- Data-link layer takes the packets from the Network Layer and encapsulates them into frames by adding header and trailer.
- Frame Size:
    - fixed-size framing: the size of the frame is fixed.
    - variable-size framing: the size of each frame varies. So a mechanism is needed to define the end of one frame and the beginning of the next. Eg: Local area networks. Two approaches are used: a character-oriented approach and a bit-oriented approach.

i) Character-Oriented (or byte-oriented) Framing:

- Data are 8-bit characters
- To separate one frame from the next, an 8-bit (1-byte) flag is added at the beginning and the end of a frame.

- The flag is a pattern which signals the start or end of a frame.



- When this pattern appears in the data, the receiver may misinterpret it as the end of the frame.
- To fix this problem, a **byte-stuffing strategy** was added to character-oriented framing.
- In byte stuffing (or character stuffing), a special byte is added to the data section of the frame when there is a character with the same pattern as the flag.
- The data section is stuffed with an extra byte. This byte is usually called the escape character (ESC) and has a predefined bit pattern.
- Whenever the receiver encounters the ESC character, it removes it from the data section and treats the next character as data, not as a delimiting flag.

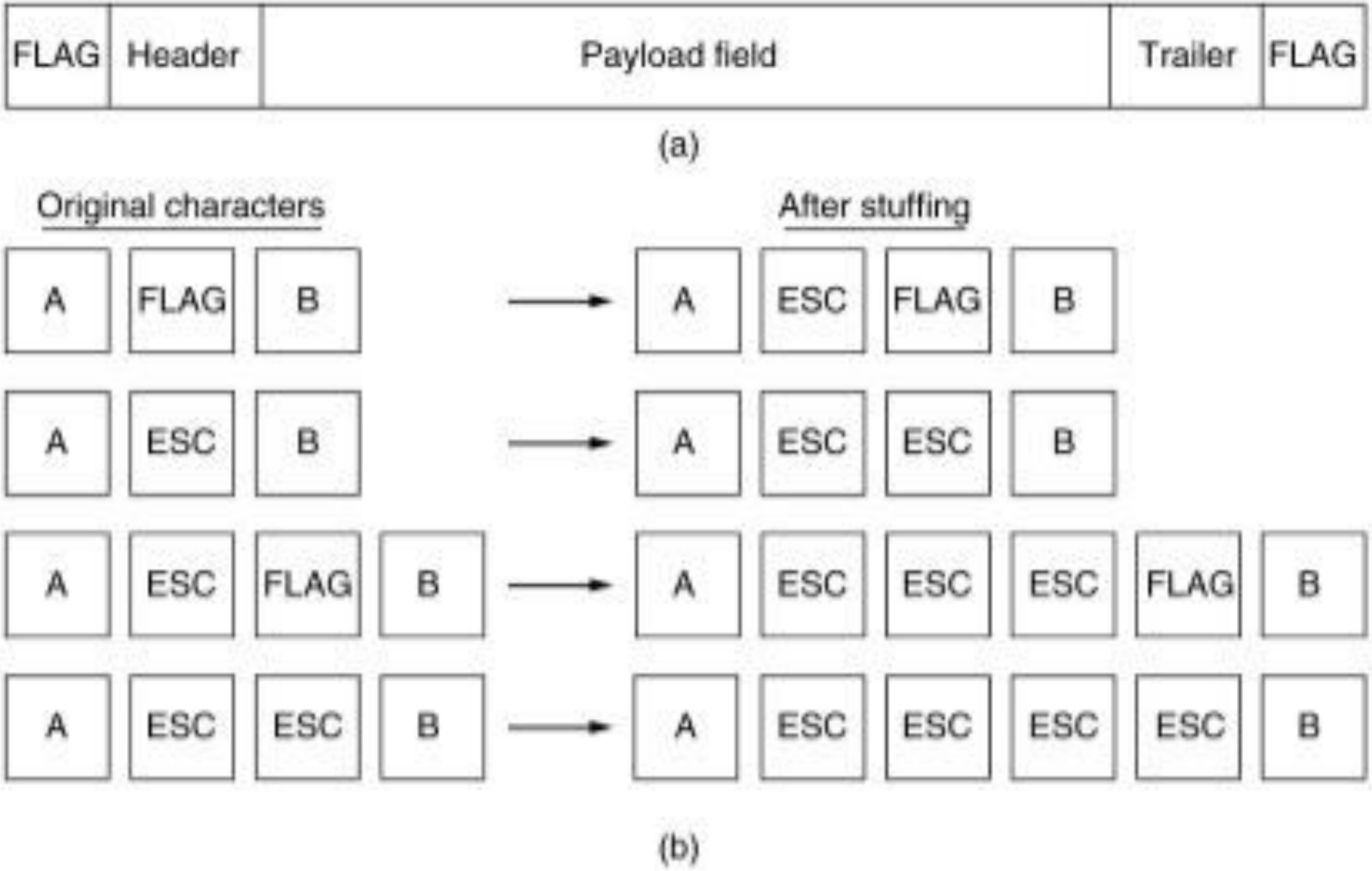
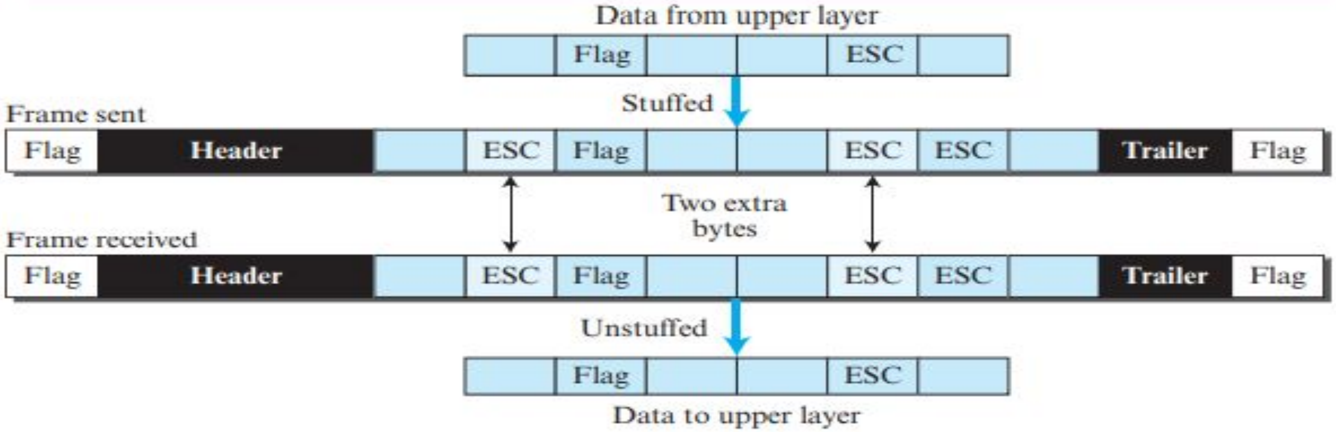


Fig2: Framing with Byte stuffing

- the escape characters that are part of the text must also be marked by another escape character.
- In other words, if the escape character is part of the text, an extra one is added to show that the second one is part of the text.

Figure *Byte stuffing and unstuffing*

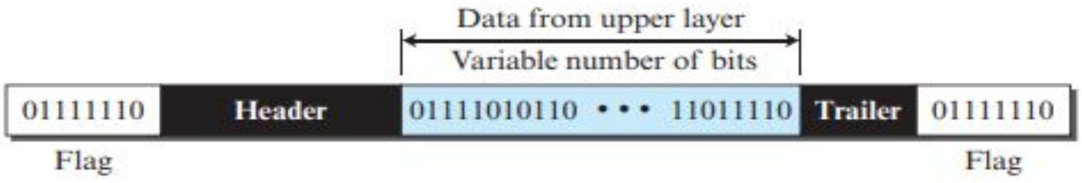


Byte stuffing is the process of adding one extra byte whenever there is a flag or escape character in the text.

ii) Bit-Oriented Framing:

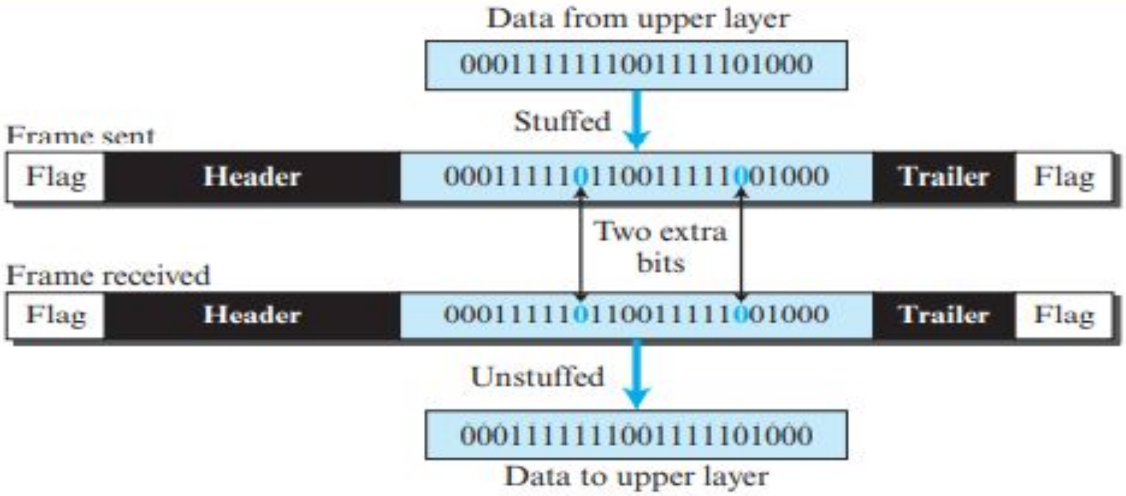
- the data section of a frame is a sequence of bits. In addition to headers (and possible trailers), a delimiter is needed to separate one frame from the other.
- Most protocols use a special 8-bit pattern flag, 01111110, as the delimiter to define the beginning and the end of the frame.
- if the flag pattern appears in the data, the receiver may misinterpret it as the end of the frame. To avoid this, bit stuffing is used, which stuffs a single bit to prevent the pattern from looking like a flag.
- In bit stuffing, if a 0 and five consecutive 1 bits are encountered, an extra 0 is added. This extra stuffed bit is eventually removed from the data by the receiver.
-

Figure *A frame in a bit-oriented protocol*



Bit stuffing is the process of adding one extra 0 whenever five consecutive 1s follow a 0 in the data, so that the receiver does not mistake the pattern 0111110 for a flag.

Figure *Bit stuffing and unstuffing*



- The real flag 01111110 is not stuffed by the sender and is recognized by the receiver

Error Detection and Correction

Error Detection and Correction

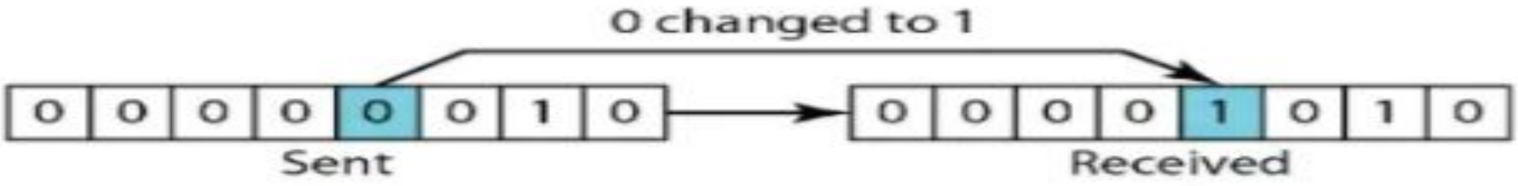
- At the data-link layer, if a frame is corrupted between the two nodes, it needs to be corrected before it continues its journey to other nodes.

Types of Errors:

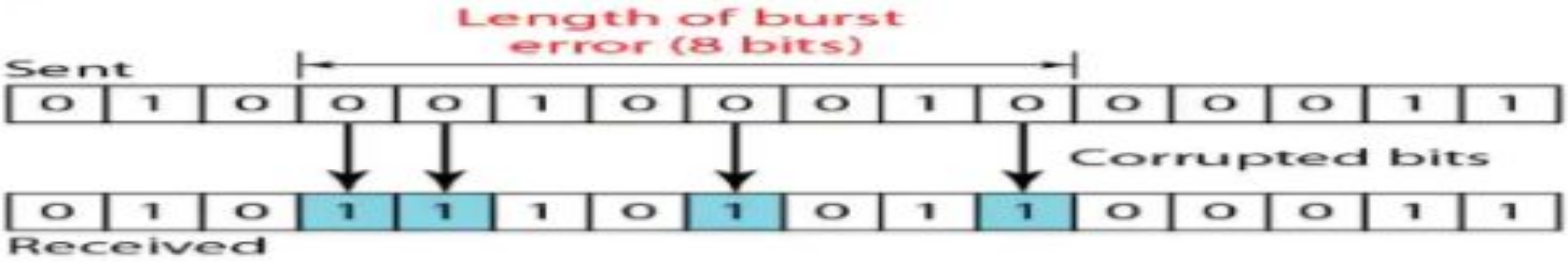
- single-bit error: only 1 bit of a given data unit is changed from 1 to 0 or from 0 to 1.
- burst error: 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1.
- The number of bits affected depends on the data rate and duration of noise.

Types of Error   Single Bit Errors

In a single bit error, only 1 bit in the data unit changes:



Burst Errors



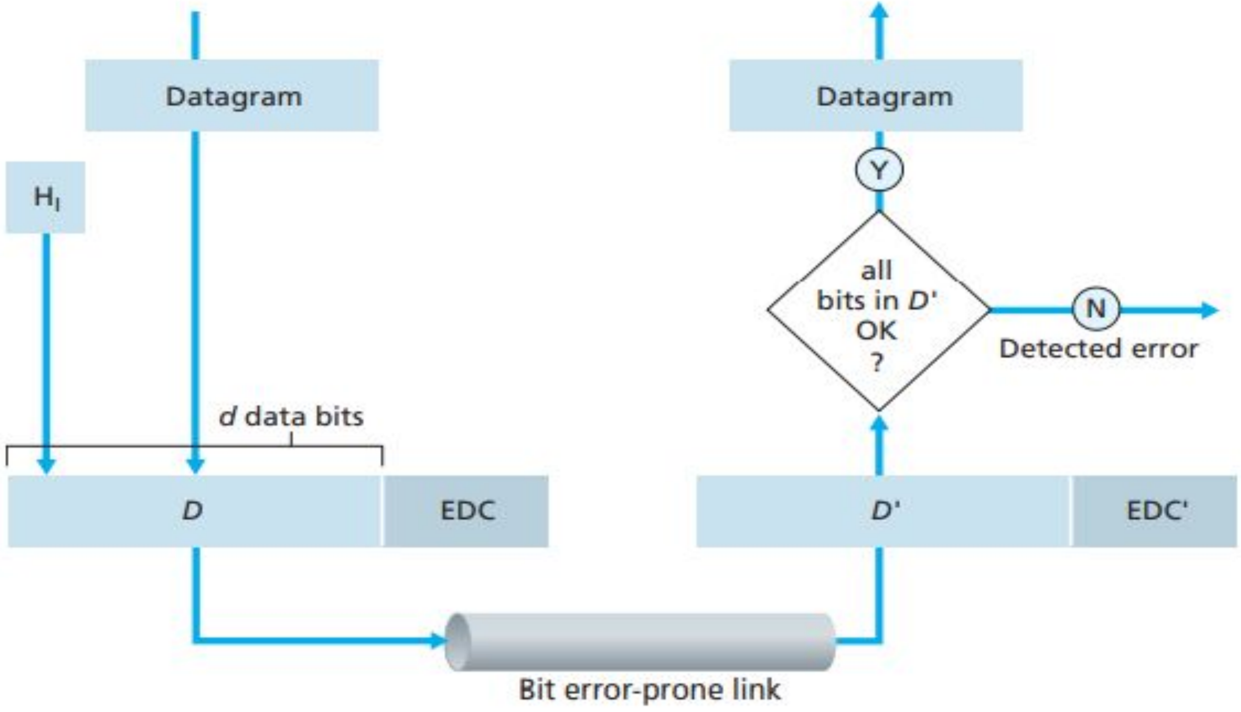
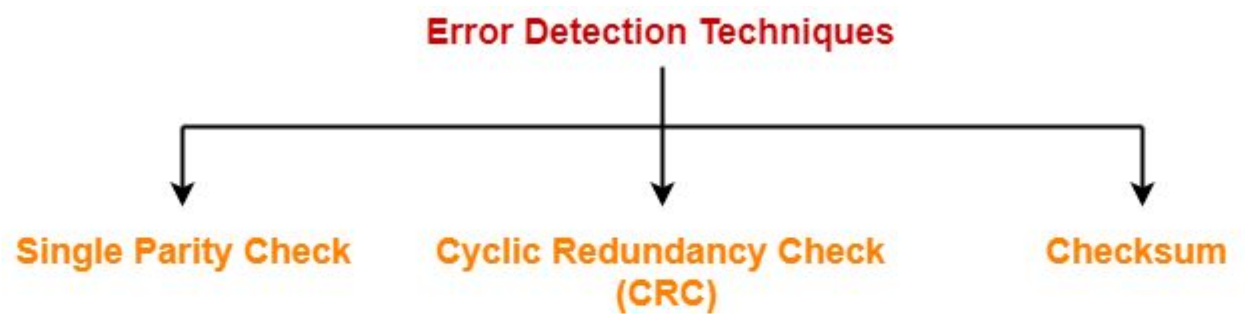


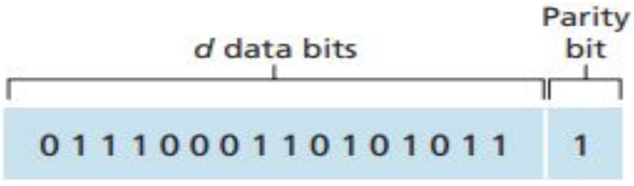
Figure ♦ Error-detection and -correction scenario

Error Detection techniques:-



i) Parity checks:

- the simplest form of error detection
- uses a single parity bit.
- Let the information to be sent, D has d bits.
- Even parity:
    - the sender simply includes one additional bit and chooses its value such that the total number of 1s in the d + 1 bits (the original information plus a parity bit) is even.



One-bit even parity

- odd parity:
  - the parity bit value is chosen such that there is an odd number of 1s.
- Example:

i) if the original data is 1010001,

For even parity; parity bit is 1 and the code word is 10100011

For odd parity; parity bit is 0 and the code word is 10100010

ii) if the original data is 1101001,

For even parity; parity bit is 0 and the code word is 11010010

For odd parity; parity bit is 1 and the code word is 11010011

- This technique is guaranteed to detect an odd number of bit errors (one, three, five and so on).
- But this technique can not detect an even number of bit errors (two, four, six and so on).

ii) Checksum:

- Sender:
    - the message is first divided into m-bit units
    - All the m bit units are then added.
    - Then 1’s complement of the sum is taken which is the checksum value.
    - The data along with checksum is then sent to the receiver.
- Receiver:
    - the received message is first divided into m-bit units.
    - All the m bit units are then added along with the checksum value.
    - Then take the 1’s complement of the result.
    - If the result is zero, then receiver assumes that no error has occurred and accepts the data; otherwise, the receiver assumes that error has occurred and discards the data.

- Consider the data to be transmitted: 10011001111000100010010010000100
- Consider 8 bit checksum is used.
- Sender:
  - Divide into 8 bit units:

    10011001 11100010 00100100 10000100

  - Take the sum of these units (1000100011). The sum is 10 bits so extra 2 bits are wrapped around. Thus we get 00100101 (00100011 + 10)
  - Take 1’s complement: 11011010 which is the checksum value.

- Receiver:

  - Take the sum of all units and checksum; Sum of all units + Checksum value = 00100101 + 11011010 = 11111111
  - Take 1’s complement; we get all 0’s. Thus the receiver assumes that no error has occurred and accepts the data.

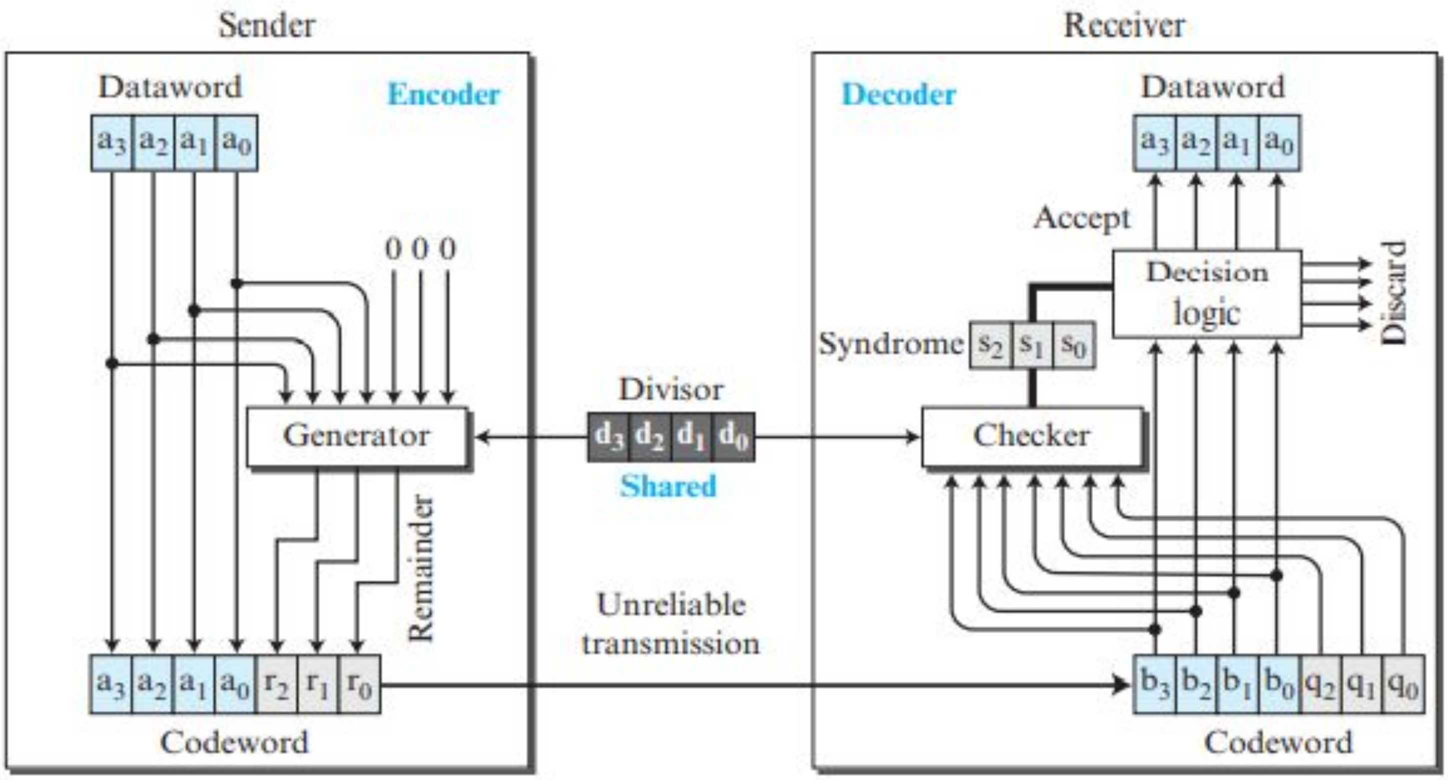
iii) Cyclic Redundancy Check(CRC) codes:

- also known as polynomial codes (includes polynomials and binary division)
- CRC uses a polynomial (as divisor) which is available on both sender and receiver side.
- The polynomial is then represented as a bit pattern.
- Eg: if the polynomial is $x^3 + x + 1$ then the bit pattern is determined as follows:

  The power of each term gives the position of the bit and the coefficient gives the value of the bit.

  - ie., $x^3 + x + 1$ can be written as $1x^3 + 0x^2 + 1x^1 + 1x^0$
  - Its bit pattern is 1011.

Figure *CRC encoder and decoder*



Sender:

- Let the dataword be k bits and the codeword be n bits.
- First of all, (n-k) 0’s are appended to the dataword to make it n bits.
- The n-bit result is then fed into the generator.
- The generator uses a divisor of size $n - k + 1$.
- The generator divides the augmented dataword by the divisor (modulo-2 division).

  [modulo-2 division is same as division except that instead of subtraction, XOR is done]

- The quotient of the division is discarded; the remainder ($r_2r_1r_0$) is called CRC and it is appended to the dataword to create the codeword(n bits).
- The generated codeword is then transmitted to the receiver.

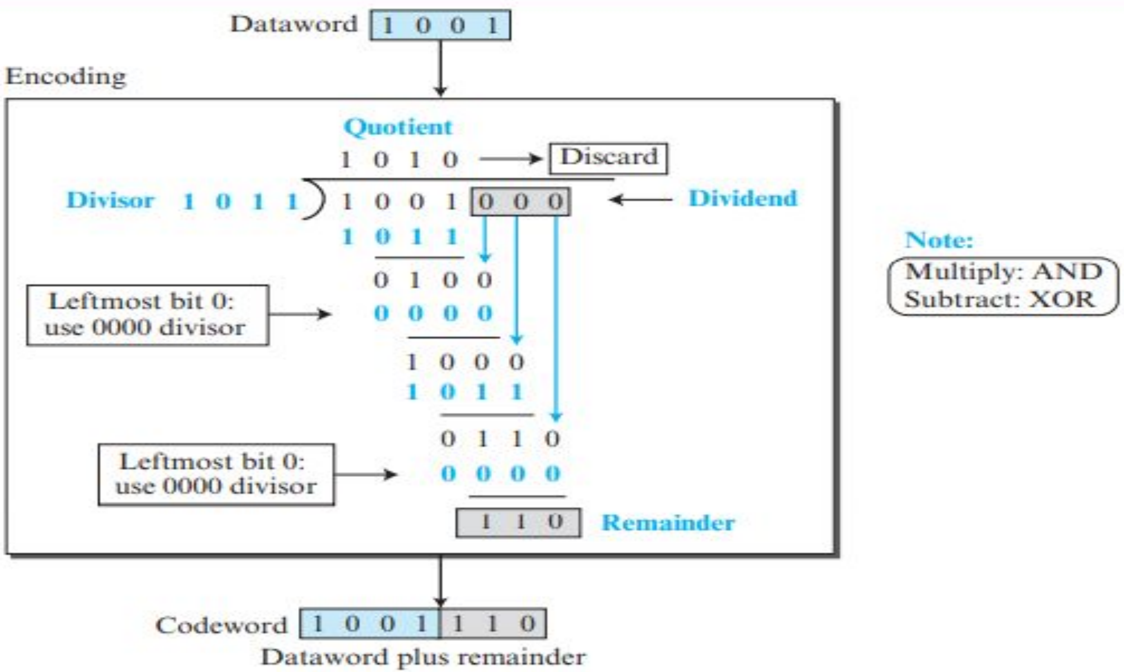
Receiver:

- The received codeword is sent to the checker(a replica of the generator).
- The checker returns a remainder of (n-k) bits, which is fed to the decision logic analyzer.
- If all (n-k) bits are 0s, then k leftmost bits of the codeword are accepted as the dataword (interpreted as no error); otherwise, the k leftmost bits are discarded (error).

Example:

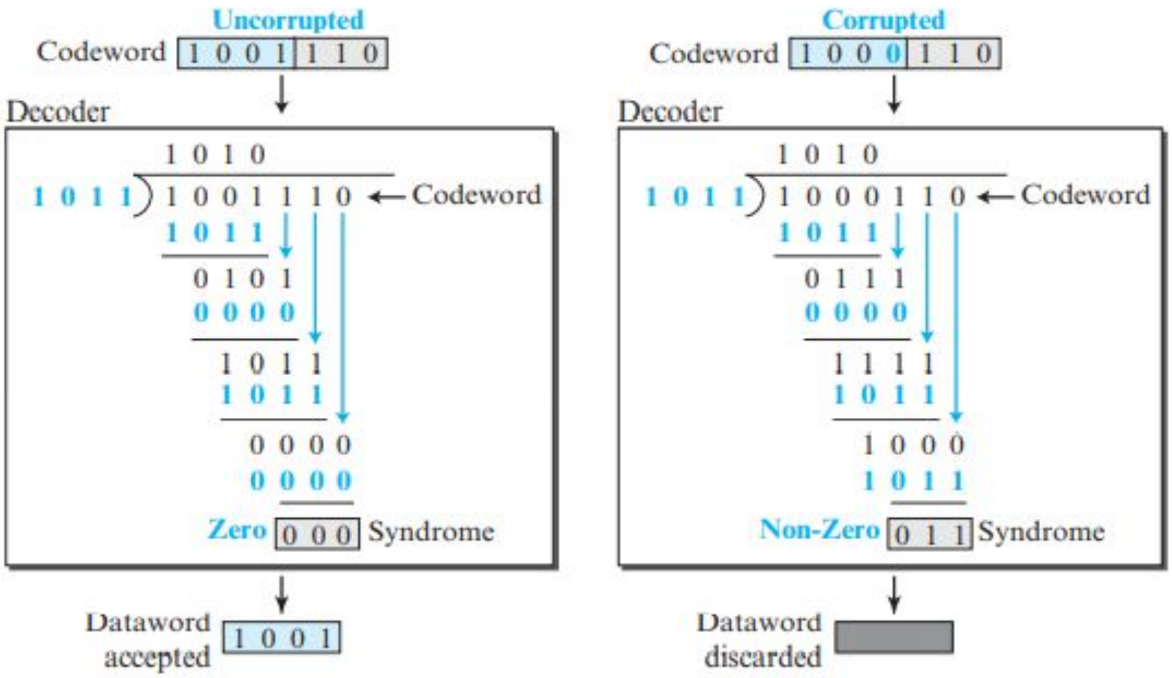
Sender:

Figure 5.13 *Division in CRC encoder*



Receiver:

Figure 5.14 *Division in the CRC decoder for two cases*



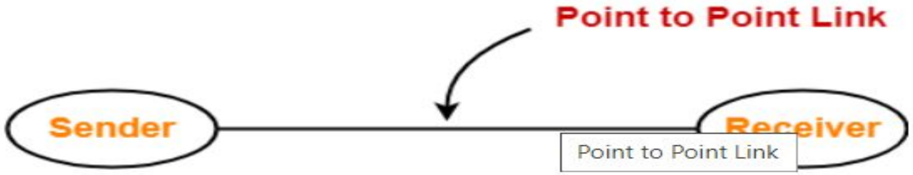
MULTIPLE ACCESS LINKS AND PROTOCOLS

Types of network links

- Communication links enable the devices to communicate with each other.
- The types of links are:

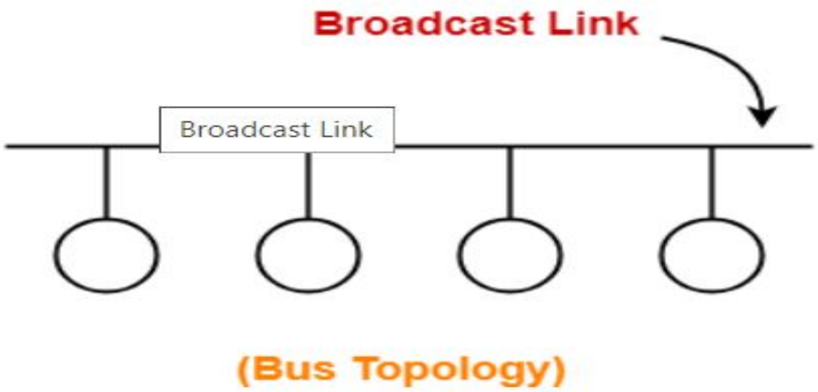
i) Point to point links:

- is a dedicated link that exists between the two stations.
- the entire capacity of the link is available only for these 2 stations.
- The link-layer protocols for point-to-point links: the point-to-point protocol (PPP) and high-level data link control (HDLC)



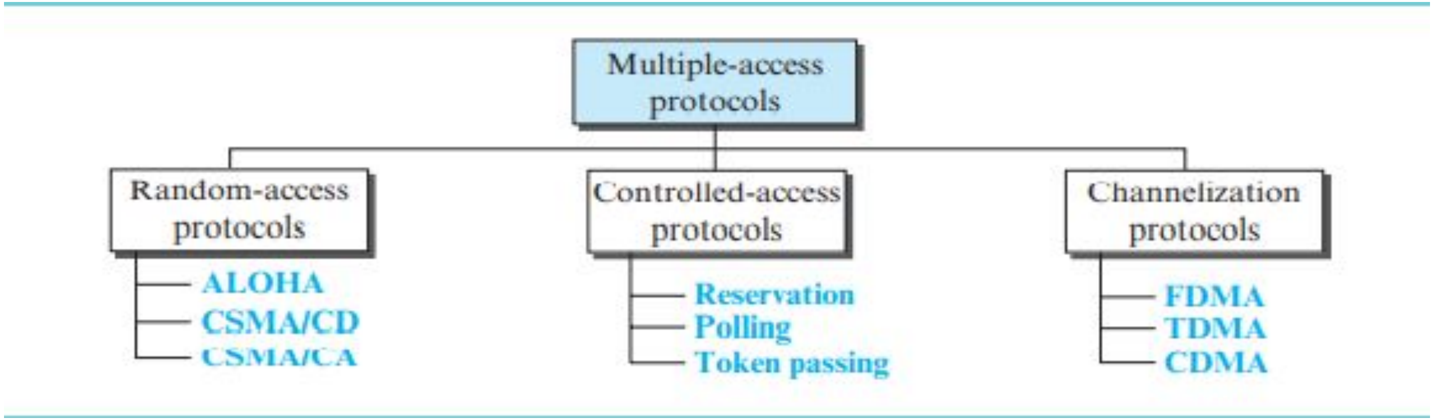
ii) Broadcast links:

- is a common link to which multiple stations are connected.
- The capacity of the link is shared among these stations.
- Eg: Ethernet and wireless LANs



MULTIPLE ACCESS PROTOCOLS

- When nodes or stations are connected via a common link, called a multipoint or broadcast link, we need a multiple-access protocol to coordinate access to the link.
- Such protocols are needed to ensure that each node gets access to the link, to prevent any collision between nodes and to handle the collision if it happens.
-



Random Access Protocols

- In random-access or contention methods, no station is superior to another station and none is assigned the control over another.
- Each station has to take a decision based on a procedure before transmitting data each time.
- This decision depends on the state of the medium (idle or busy).
- There is no scheduled time for a station to transmit. Transmission is random among the stations. That is why these methods are called **random access**.
- no rules specify which station should send next. Stations compete with one another to access the medium. That is why these methods are also called **contention methods.**
- Whenever more than one station tries to send data, it will cause collision and thereby corruption/loss of data.

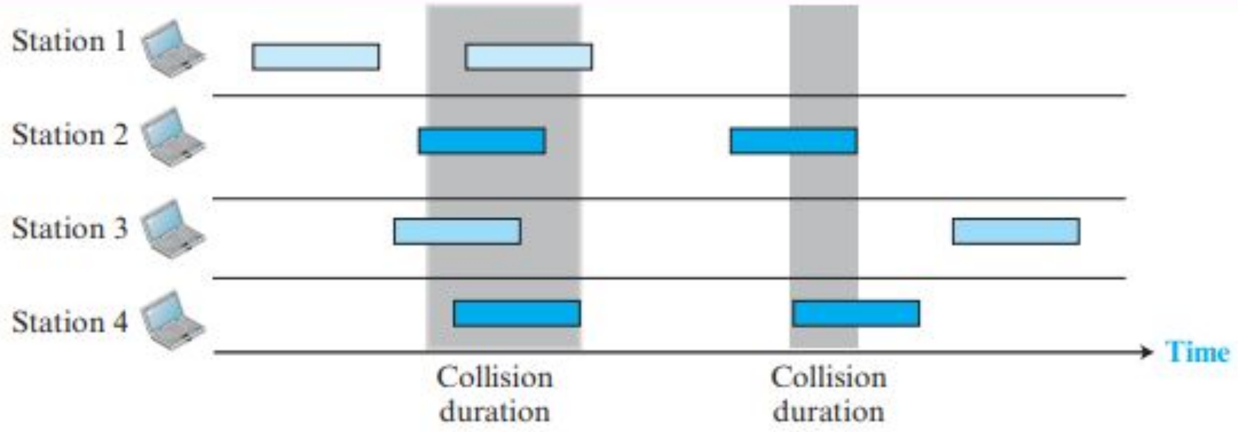
- So each station follows a procedure that answers the following questions:
    - When can the station access the medium?
    - What can the station do if the medium is busy?
    - How can the station determine the success or failure of the transmission?
    - What can the station do if there is an access conflict?

- The random-access methods are
    - ALOHA: used a very simple procedure called multiple access (MA)
    - Carrier sense multiple access (CSMA): each station has to sense the medium before transmitting
    - Carrier sense multiple access with collision detection (CSMA/CD): it tells the station what to do when a collision is detected
    - Carrier sense multiple access with collision avoidance (CSMA/CA): it tries to avoid the collision.

ALOHA

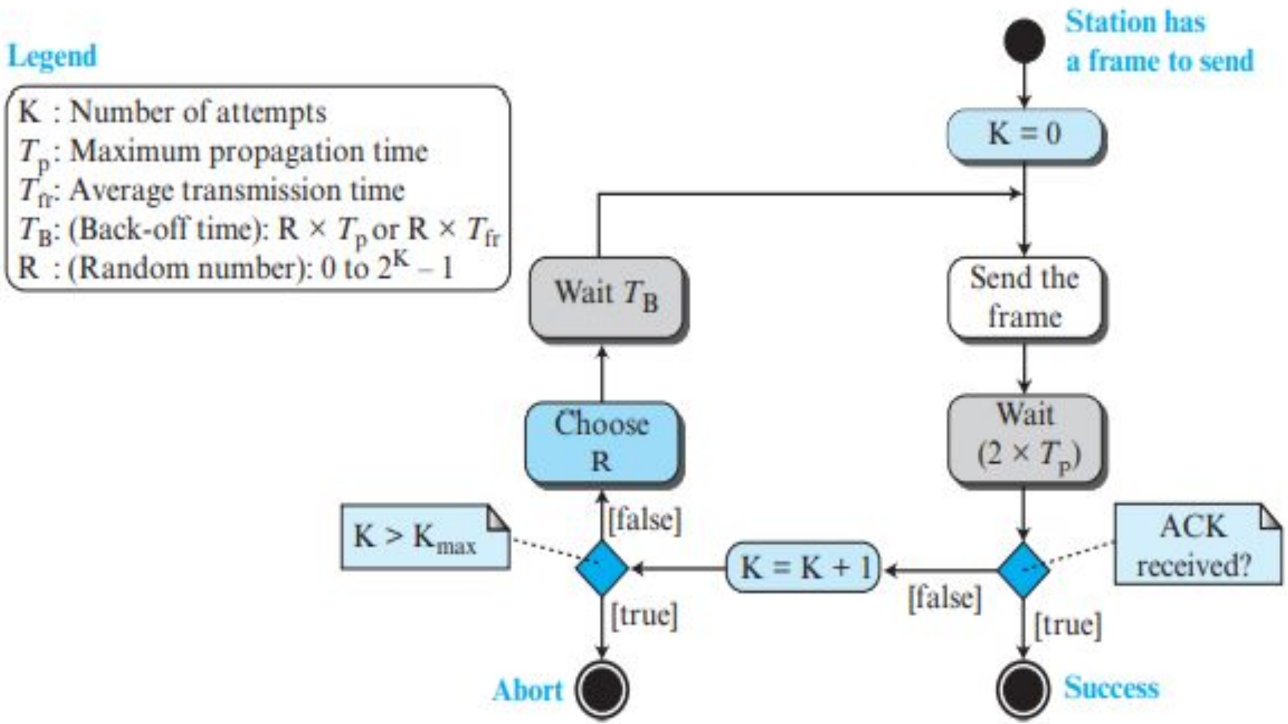
- The original ALOHA protocol is called pure ALOHA.
- each station sends a frame whenever it has a frame to send (multiple access).
- But there is a chance of collision between frames from different stations.
-

Figure Frames in a pure ALOHA network



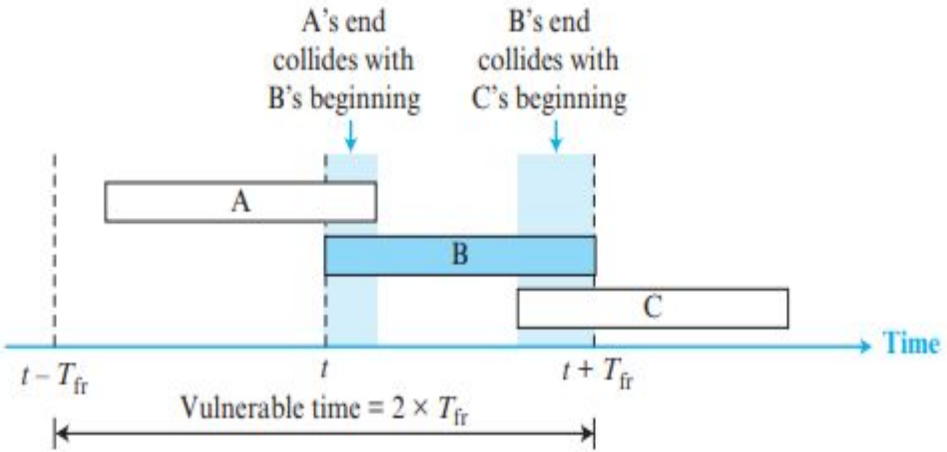
- The pure ALOHA protocol relies on acknowledgments from the receiver.
- If the acknowledgment does not arrive within a particular period of time, the station resends the frame.
- A collision involves two or more stations.
- Pure ALOHA dictates that when the time-out period passes, each station waits a random amount of time before resending its frame. The randomness will help avoid more collisions.
- That time is called the back-off time T_B.
- Pure ALOHA has a second method to prevent congesting the channel with retransmitted frames.
- After a maximum number of retransmission attempts Kmax, a station must give up and try later.
-

Figure *Procedure for pure ALOHA protocol*



- Vulnerable time:

Figure Vulnerable time for pure ALOHA protocol



Pure ALOHA vulnerable time = $2 \times T_{fr}$

The throughput for pure ALOHA is $S = G \times e^{-2G}$.
The maximum throughput $S_{max} = 1/(2e) = 0.184$ when $G = (1/2)$.

- where G is the number of stations that wishes to transmit at the same time.

Pure Aloha

efficiency, $S = G \times e^{-2G}$

to find max efficiency $\rightarrow$ $\frac{\partial S}{\partial G} = G \times e^{-2G}(-2) + e^{-2G}(1)$

$= e^{-2G}(-2G+1) = 0$

$-2G+1=0$

$G = 1/2$

Ie., when half of the stations transmit at the same time, collisions will be reduced & efficiency will be max.

$S_{max} = \frac{1}{2} \times e^{-\cancel{2} \times 1/2}$

$= \frac{1}{2} \times e^{-1} = \cancel{\frac{2.71}{2}} = \frac{1}{2e} = \frac{1}{2 \times 2.71}$

$= 0.184$

$= 18.4\%$

SLOTTED ALOHA

- The time of the shared channel is divided into discrete time intervals called time slots.
- Stations can send data only at the beginning of these slots.
- if a station misses a time slot, it must wait until the beginning of the next time slot.

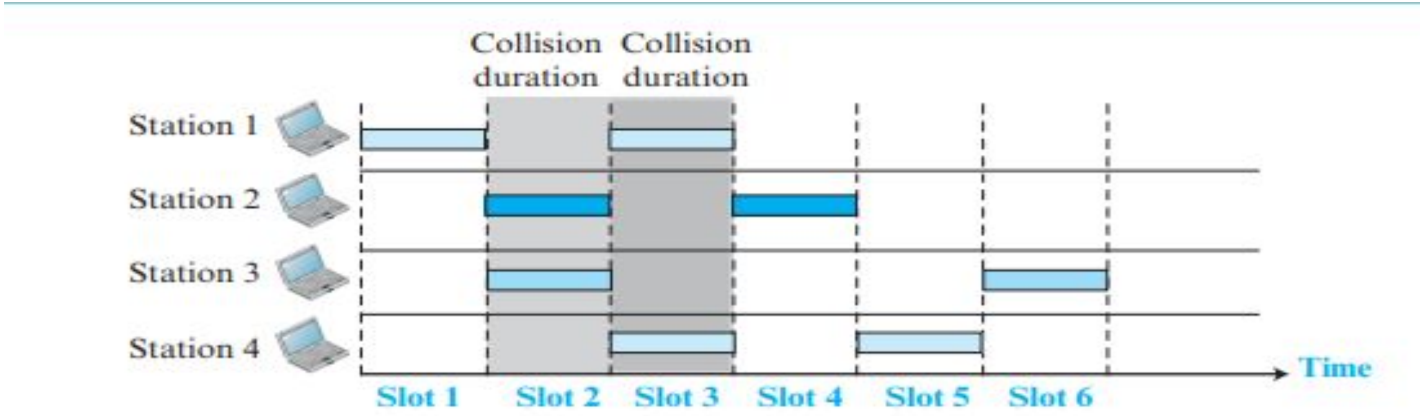
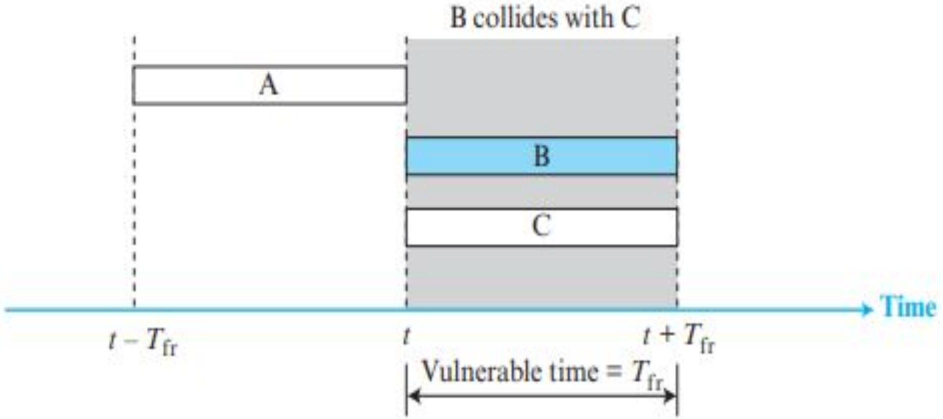


Figure *Vulnerable time for slotted ALOHA protocol*



Slotted ALOHA vulnerable time = T_{fr}

The throughput for slotted ALOHA is $S = G \times e^{-G}$.

The maximum throughput $S_{max} = 0.368$ when $G = 1$.

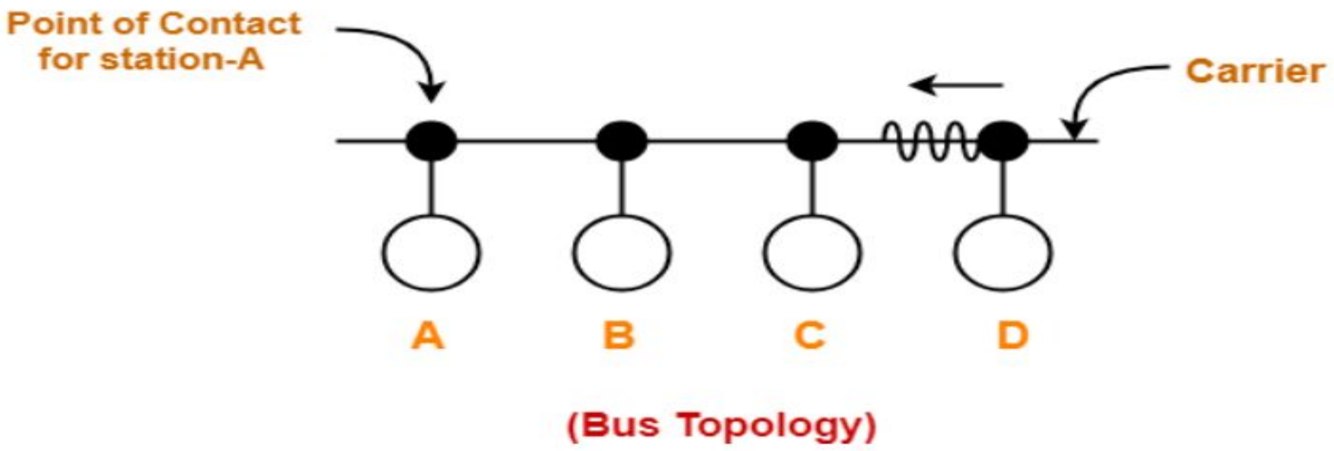
Key	Pure Aloha	Slotted Aloha
Time Slot	In Pure Aloha, any station can transmit data at any time.	In Slotted Aloha, any station can transmit data only at the beginning of a time slot.
Time	In Pure Aloha, time is continuous and is not globally synchronized.	In Slotted Aloha, time is discrete and is globally synchronized.
Vulnerable time	The vulnerable time or susceptible time in Pure Aloha is equal to (2Ã□Tt).	In Slotted Aloha, the vulnerable time is equal to (Tt).
Probability	The probability of successful transmission of a data packet $S=G\times e^{-2G}$	The probability of successful transmission of data packet $S=G\times e^{-G}$
Maximum efficiency	Maximum efficiency = 18.4%.	Maximum efficiency = 36.8%.
Number of collisions	Does not reduce the number of collisions.	Slotted Aloha reduces the number of collisions to half, thus doubles the efficiency.

Carrier Sense Multiple Access (CSMA):

- To minimize the chance of collision and, therefore, increase the performance, the CSMA method was developed.
- Carrier sense multiple access (CSMA) requires that each station first listen to the medium (or check the state of the medium) before sending.

Principle of CSMA: **“sense before transmit” or “listen before talk.”**

- Carrier busy: some transmission is going on
- Carrier idle: currently no transmission at all
- If the station finds the carrier free, it starts transmitting its data packet otherwise not.

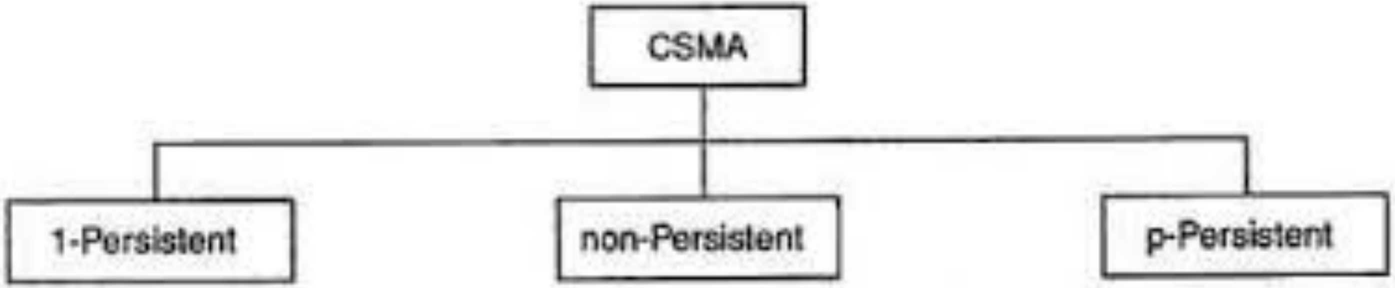


- Each station can sense the carrier only at its point of contact with the carrier, not the entire carrier.
- The possibility of collision still exists because of propagation delay.
- Vulnerable time=propagation time

Working Principle

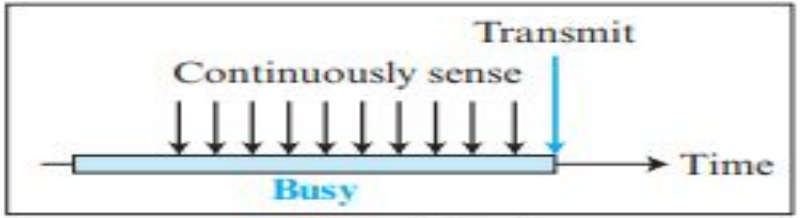
- When a station has frames to transmit, it attempts to detect presence of the carrier signal from the other nodes connected to the shared channel.
- If a carrier signal is detected, it implies that a transmission is in progress.
- The station waits till the ongoing transmission executes to completion, and then initiates its own transmission. Generally, transmissions by the node are received by all other nodes connected to the channel.
- Since, the nodes detect for a transmission before sending their own frames,collision of frames is reduced. However, if two nodes detect an idle channel at the same time, they may simultaneously initiate transmission.
- This would cause the frames to garble resulting in a collision.

Persistence Methods:

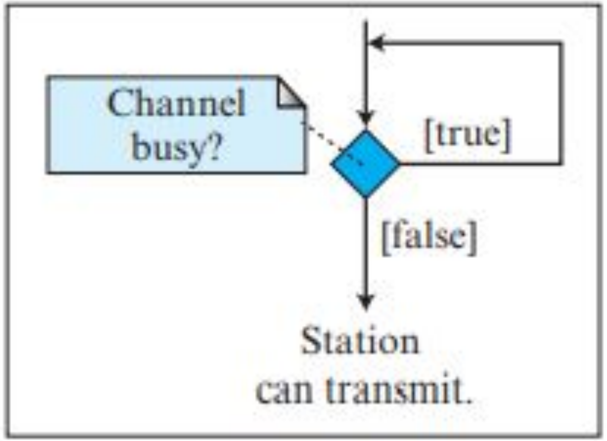


Types of CSMA

- **1-Persistent:**
    - The station first of all sense the channel.
    - If it is idle, the station sends its frame immediately (with probability 1).
    - This method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately.

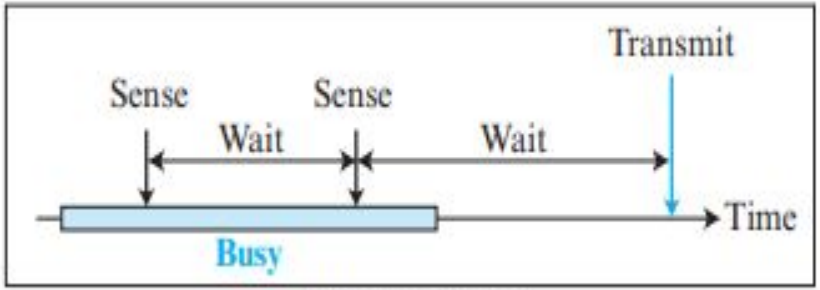


a. 1-persistent

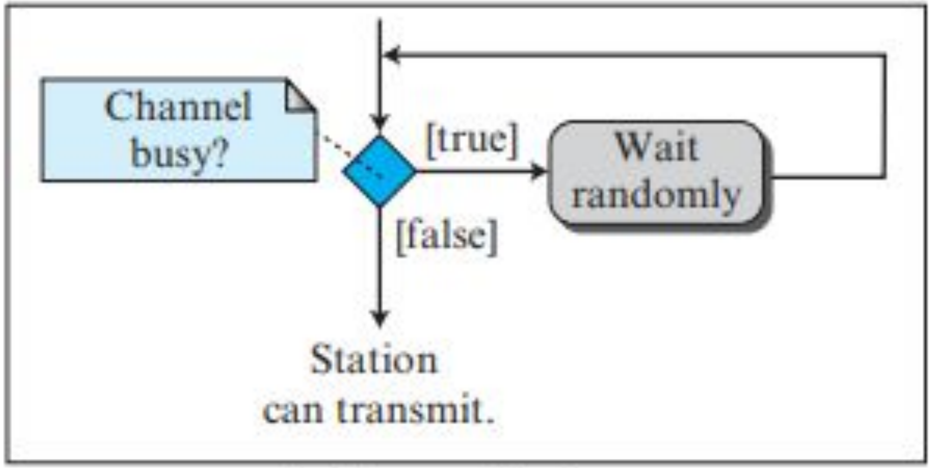


a. 1-persistent

- **Non-persistent:**
  - The station first of all sense the channel.
  - If the line is idle, it sends immediately; otherwise, it waits a random amount of time and then senses the line again.
    - It reduces the chance of collision but it also reduces the efficiency of the network because the medium remains idle when there may be stations with frames to send.



b. Nonpersistent



b. Nonpersistent

iii) p-persistent:

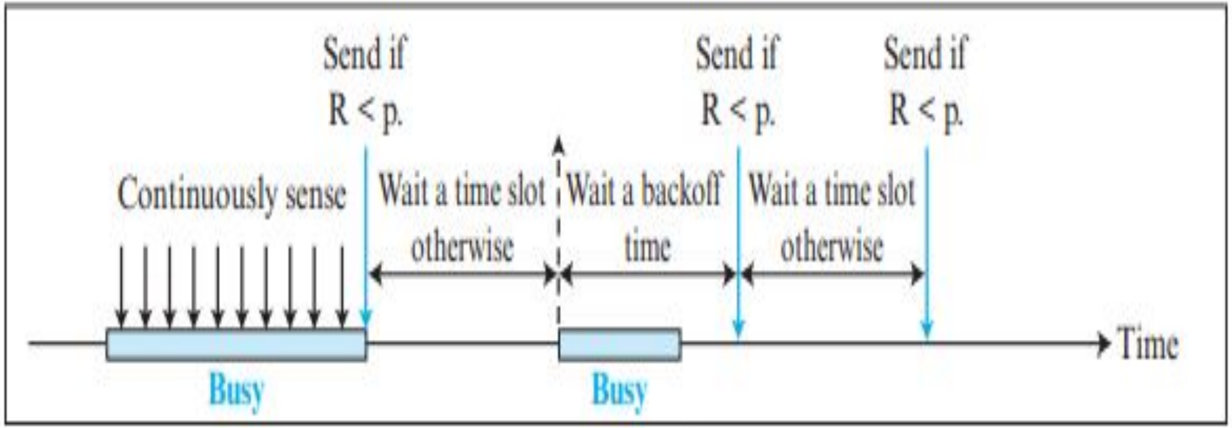
- It is a combination of 1-persistent and non-persistent methods.
- It reduces the chance of collision and improves efficiency.
- After the station finds the line idle it follows these steps:

1. With probability p, the station sends its frame.

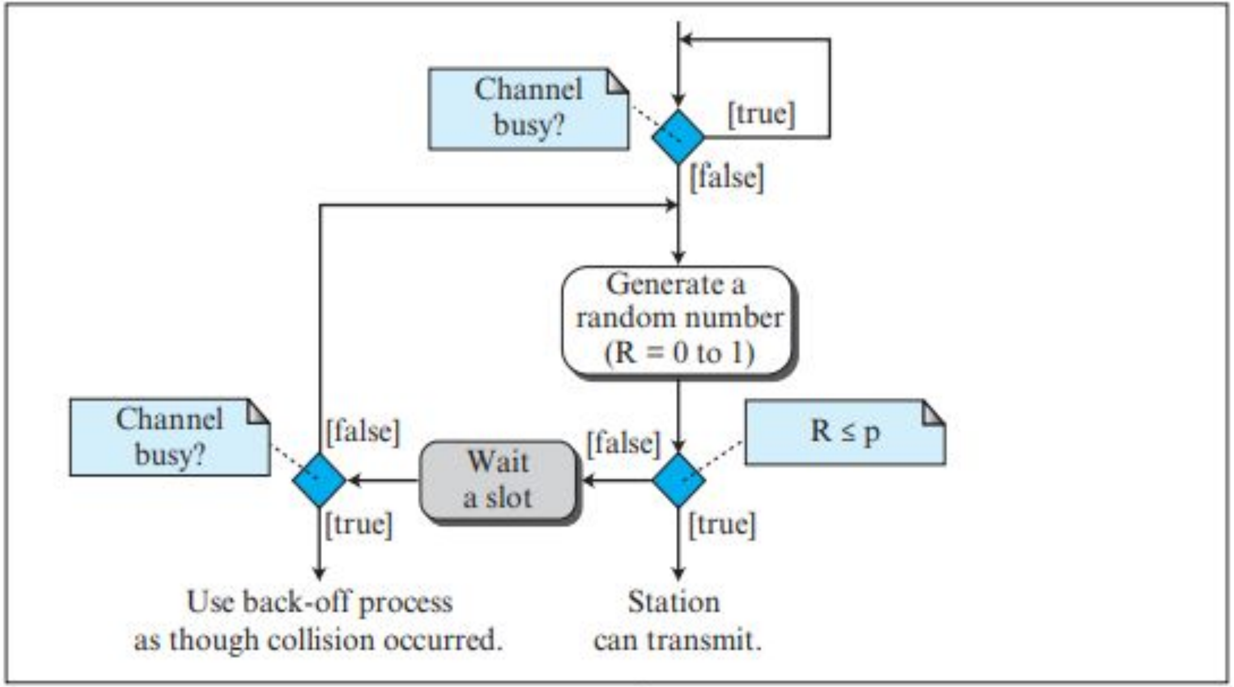
2. With probability q = 1 − p, the station waits for the beginning of the next time slot and checks the line again.

a. If the line is idle, it goes to step 1.

b. If the line is busy, it acts as though a collision has occurred and uses the backoff procedure.



c. *p*-persistent



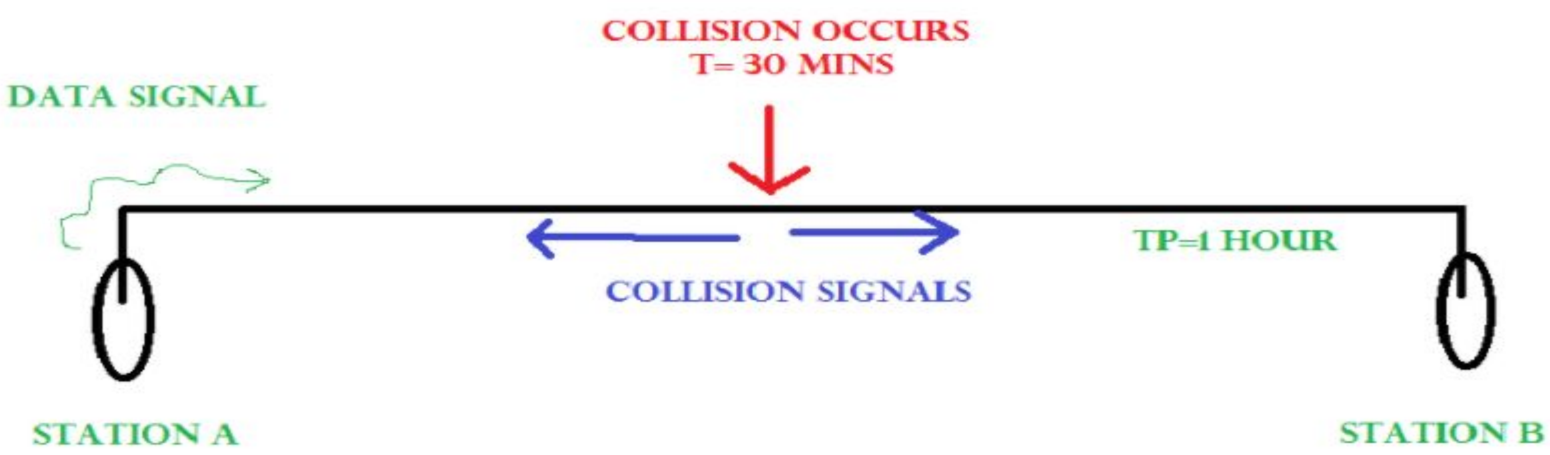
c. *p*-persistent

CSMA/CD:

- Carrier sense multiple access with collision detection (CSMA/CD) augments the algorithm to handle the collision.
- widely used in Early Ethernet technology/LANs.

Working:

- When a station wants to transmit, it listens to the transmission medium.
- Check if the transmission link is idle?
  - If it senses that the carrier is free and there are no collisions, it sends the data. Otherwise it refrains from sending data.
- Transmit the data & check for collisions.
  - CSMA/CD does not use ‘acknowledgement’ system.
  - It checks for the successful and unsuccessful transmissions through collision signals.
  - During transmission, if collision signal is received by the node, transmission is stopped.
  - The station then transmits a jam signal onto the link to alert other stations about the collision and waits for random time interval before it re-sends the frame.
  - After some random time, it again attempts to transfer the data and repeats above process.
- If no collision was detected in propagation, the sender completes its frame transmission.



Started transmn at 10 am

Started transmn at 10 am

Suppose Collision happens at 10.30 am

Collision signals reach at 11.00 am

Collision signals reach at 11.00 am

- **Transmission time (Tt) > Propagation Time (Tp) : to ensure that its our station's data that collided.**
- **The more accurate condition is …. Tt>=2Tp**

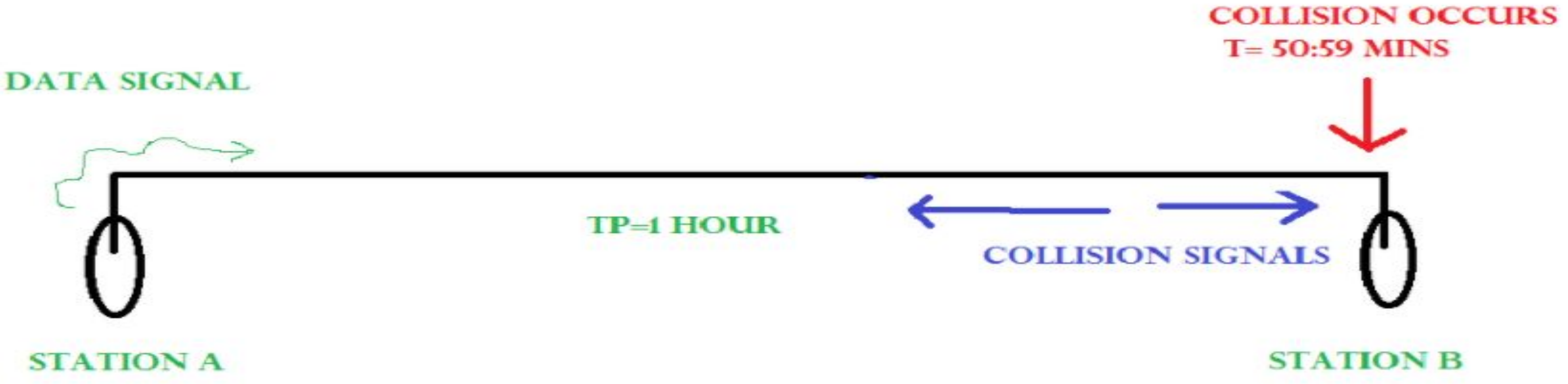
- In CSMA/CD (Carrier Sense Multiple Access with Collision Detection), ensuring that the transmitted data is still actively being sent when a collision is detected is crucial.
- This is achieved by making the transmission time (Tt) greater than or equal to twice the propagation time (Tp), or Tt >= 2*Tp.
- This guarantees that the sender can detect the collision signal (resulting from the collided data) before it finishes transmitting the data packet, allowing it to determine if its own data was involved in the collision.

Explanation:

Propagation Time (Tp):
This is the time it takes for a signal to travel from one end of the network to the other.

Transmission Time (Tt):
This is the time it takes for a station to transmit its entire frame (packet).
Collision Detection:
When two or more stations transmit data simultaneously, a collision occurs.
Why Tt >= 2Tp?*

- In the worst-case scenario, the collision signal will travel from one end of the network to the other (propagation time, Tp) and back to the sender (another propagation time, Tp).
- Therefore, the sender needs to be transmitting its data for a minimum of 2*Tp to ensure it can detect the collision signal before it finishes transmitting.
- If Tt is less than 2*Tp, the sender might finish transmitting before the collision signal reaches it, making it impossible to determine if its data was involved in the collision.



- Suppose A started transmission at 10 am.
- Suppose collision happens at 10.59.59 am
- Immediately B will stop transmission; but collision signals have to travel approximately 1 hour to reach A.
- So A will get the collision signals approximately at 12pm.
- So to detect the collision completely, **Tt >= 2 * Tp**
- Tt= length of the message(L)/bandwidth(BW)
- Therefore, L >= 2 * Tp * BW

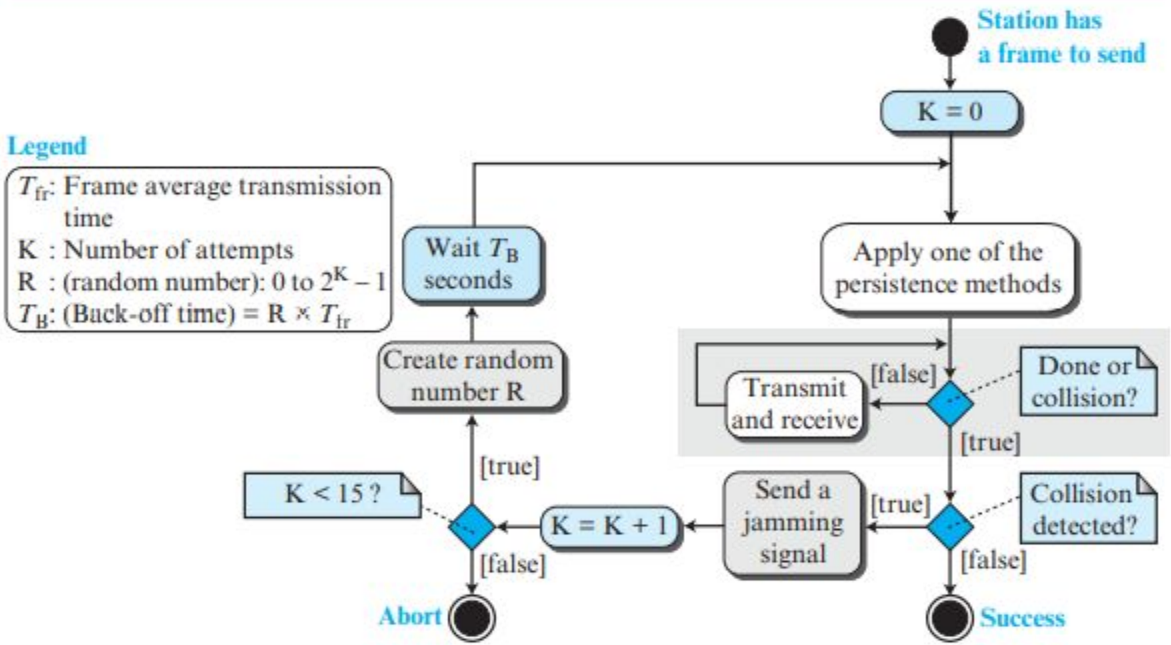
- Efficiency of CSMA/CD = 1/ (1 + 6.44a)

  Where a=Tp/Tt,

  Tp= distance/speed and

  Tt= length of the message(L)/bandwidth(BW)

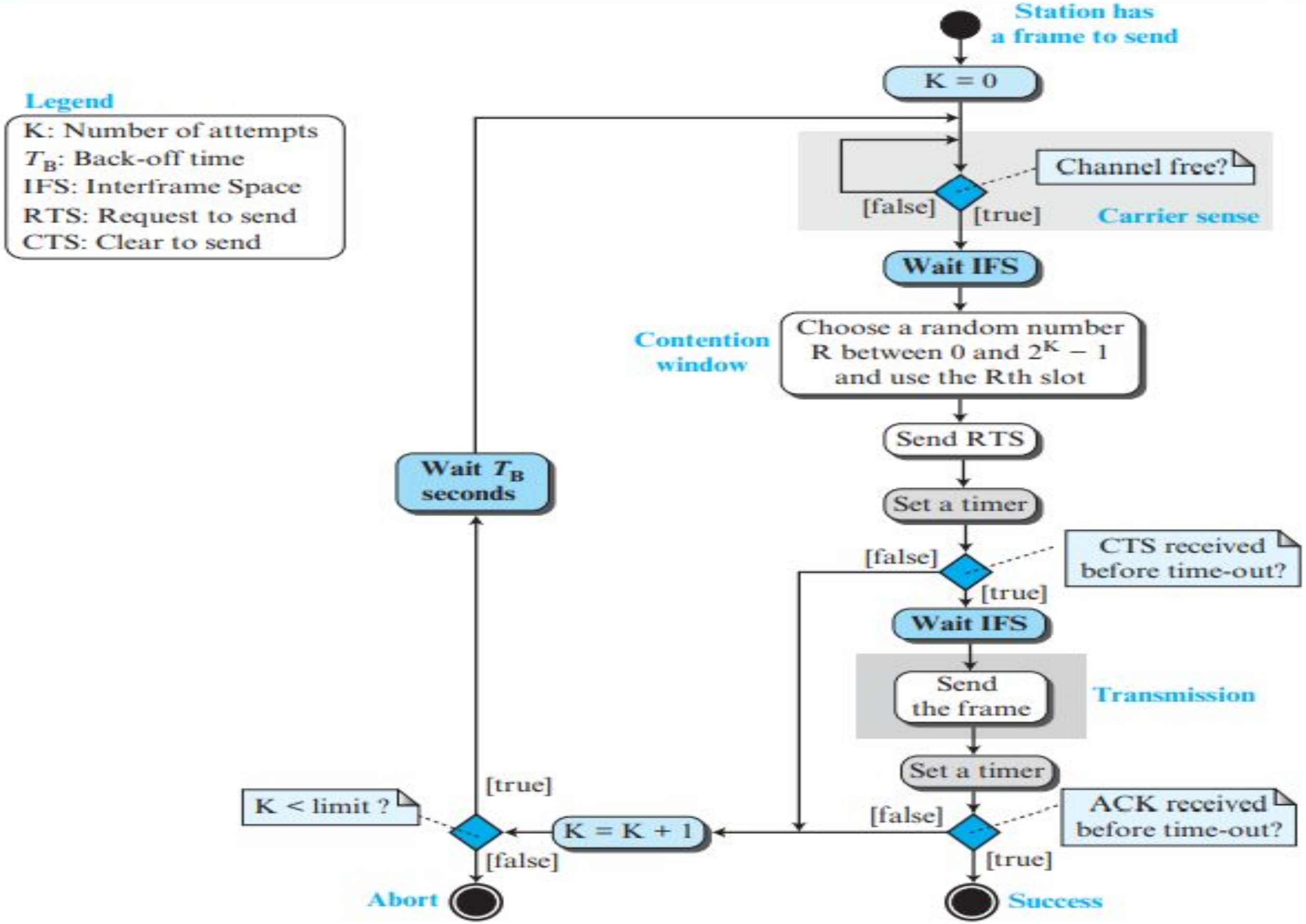
Figure *Flow diagram for the CSMA/CD*



CSMA/CA:

- Carrier Sense Multiple Access with Collision Avoidance
- It is the access method used in **wireless Lan**(wi-fi / IEEE802.11)
- It uses three types of strategies:
    - **InterFrame Space (IFS)** – When a station finds the channel busy, it waits for a period of time called IFS time. IFS can also be used to define the priority of a station or a frame. Higher the IFS lower is the priority.
    - **Contention Window** – It is the amount of time divided into slots.A station which is ready to send frames chooses random number of slots as wait time.
    - **Acknowledgements** – The positive acknowledgements and time-out timer can help guarantee a successful transmission of the frame.
-

Figure *Flow diagram of CSMA/CA*



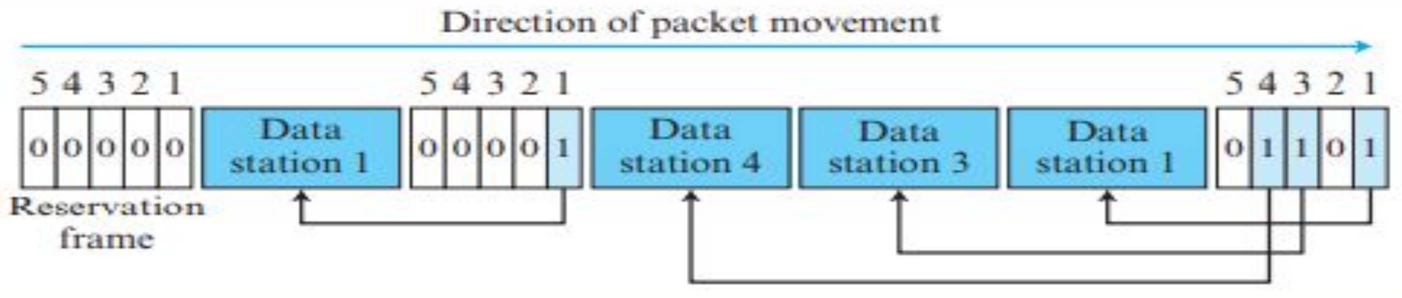
Controlled Access Protocols

- The stations consult one another to find which station has the right to send.
- A station cannot send unless it has been authorized by other stations.
- 3 controlled access methods are:
  - Reservation
  - Polling
  - Token passing

i) **Reservation:**

- A station needs to make a reservation before sending data.
- Time is divided into intervals.
- In each interval, a reservation frame precedes the data frames sent in that interval.
- If there are N stations in the system, there are exactly N reservation minislots in the reservation frame, each for a station.
- When a station needs to send a data frame, it makes a reservation in its own minislot.
- The stations that have made reservations can send their data frames after the reservation frame.

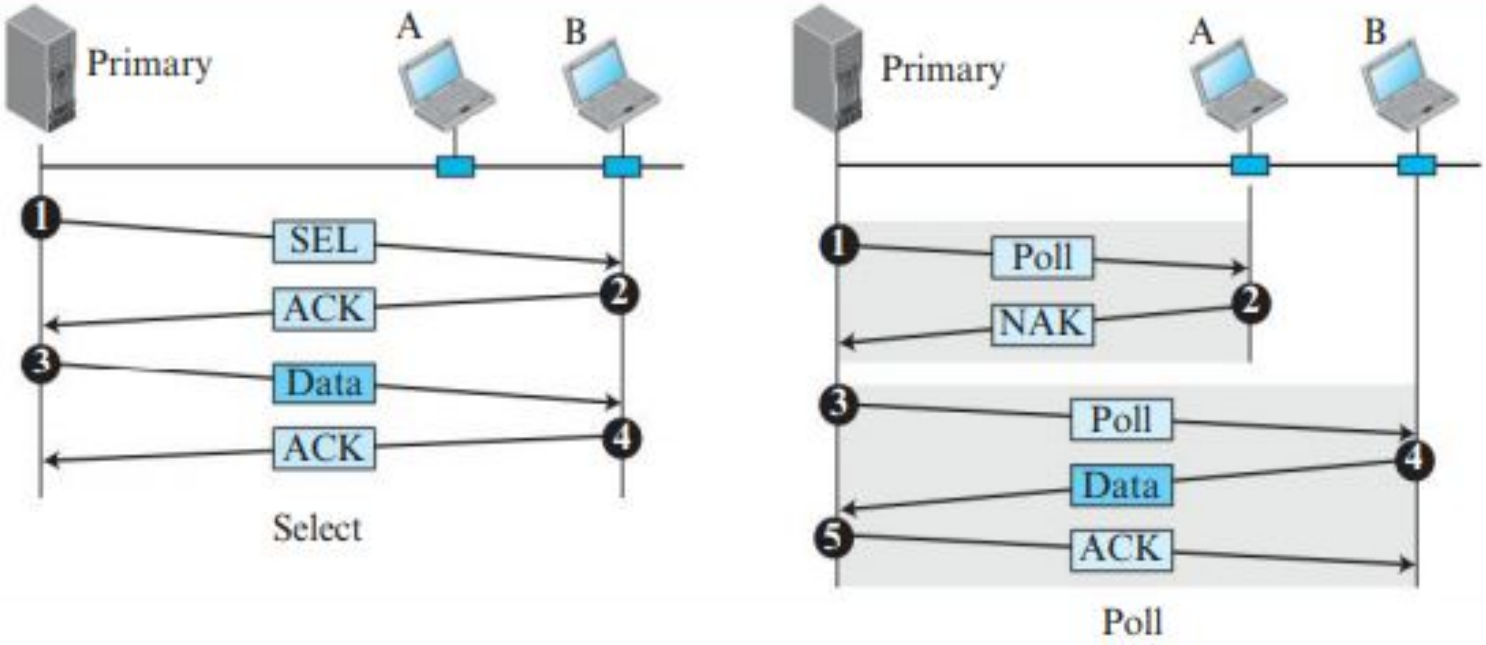
Figure 5.42 *Reservation access method*



ii) **Polling:**

- One device is designated as a primary station and the other devices are secondary stations.
- The primary device controls the link; the secondary devices follow its instructions.
- It is up to the primary device to determine which device is allowed to use the channel at a given time.
- This method uses poll and select functions to prevent collisions.
- **Select** function: It is used whenever the primary device has something to send.
- **Poll** function:
    - When the primary is ready to receive data, it must ask (poll) each device in turn if it has anything to send.
    - The primary node polls each of the nodes in a round robin fashion.
    - Other nodes respond either with a NAK if they have nothing to send or with data if they have.
- If the primary station fails, the system goes down.

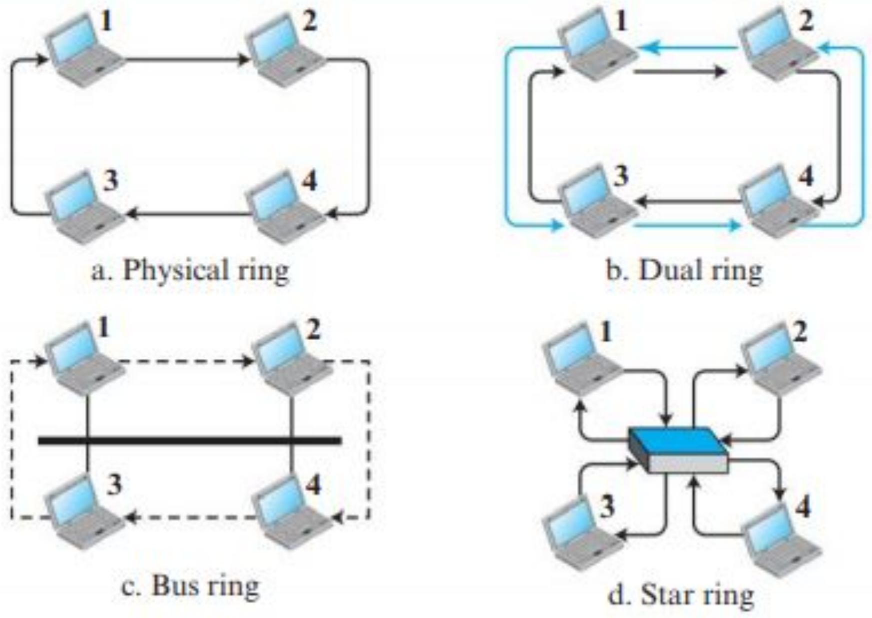
Figure *Select and poll functions in polling-access method*



iii) **Token passing:**

- the stations are connected logically to each other in form of ring and access of stations is governed by tokens.
- A token is a special bit pattern or a small message, which circulate from one station to the next in the some predefined order.
- The possession of the token gives the station the right to access the channel and send its data.
- When a station has some data to send, it waits until it receives the token from its predecessor. It then holds the token and sends its data.
- When the station has no more data to send, it releases the token, passing it to the next logical station in the ring.
- If a node fails then it may affect the entire system.

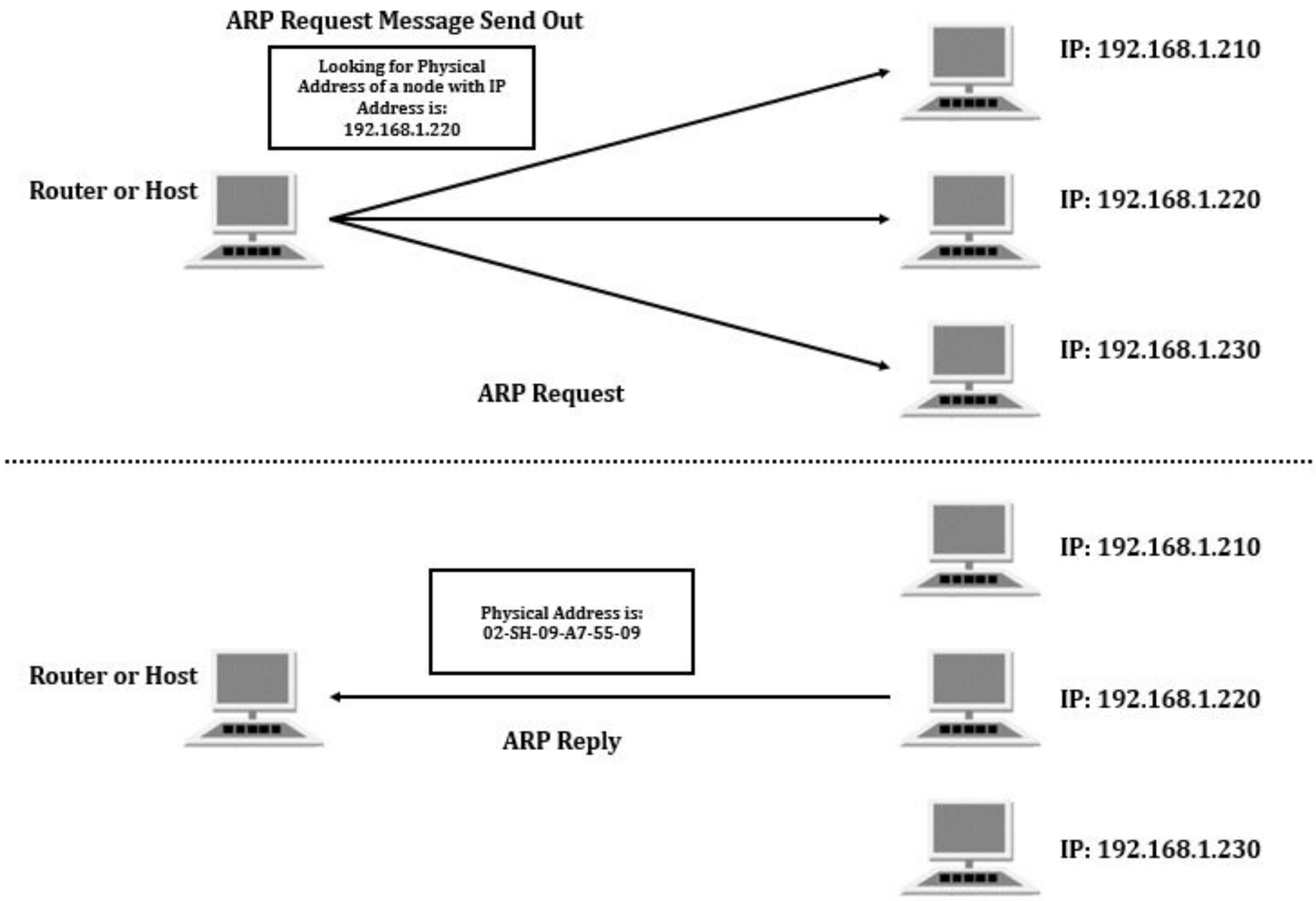
Figure *Logical ring and physical topology in token-passing access method*



ADDRESS RESOLUTION PROTOCOL (ARP)

- It belongs to the OSI data link layer (Layer 2).
- Before a host can send any data, it must know the Layer 3 (IP) and Layer 2 (MAC) addresses to correctly construct a frame.
- ARP is used to dynamically learn the MAC address of a receiver, when the IP address is known.

- Each device on the network is having an ARP table which consists of different devices on the network.
- The entries in this table are dynamically added and removed.


ARP Request Message Send Out
Looking for Physical Address of a node with IP Address is: 192.168.1.220
Router or Host
IP: 192.168.1.210
IP: 192.168.1.220
IP: 192.168.1.230
ARP Request
IP: 192.168.1.210
Physical Address is: 02-SH-09-A7-55-09
Router or Host
IP: 192.168.1.220
ARP Reply
IP: 192.168.1.230

- If a node (A) wants to find out the MAC address of another node (B), it will broadcast an ARP request message containing B’s IP address.
- All nodes on the network receives this message but only B responds to A with ARP response, giving its MAC address.
- Now A receives B’s reply and updates its ARP table with B’s MAC address.

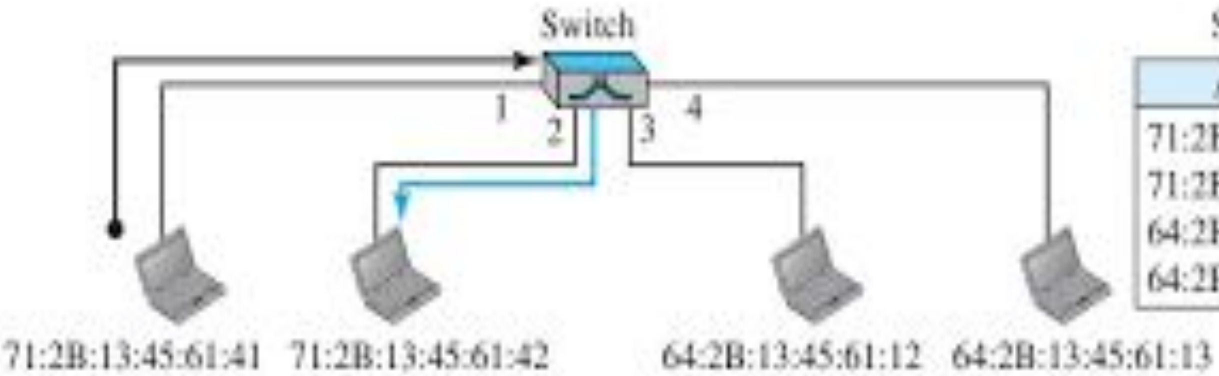
Ethernet switch

- A link-layer switch operates in both the physical and the data-link layers.
- As a physical-layer device, it regenerates the signal it receives.
- As a link-layer device, the link-layer switch can check the MAC addresses (source and destination) contained in the frame.

Filtering:

- It can check the destination address of a frame and can decide through which outgoing port the frame should be sent.

Figure 17.3 *Link-layer switch*



Switching table

Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3
64:2B:13:45:61:13	4

Transparent Switches:

- It is a switch in which the stations are completely unaware of the switch's existence.
- If a switch is added or deleted from the system, reconfiguration of the stations is unnecessary.

Learning:

- A learning switch uses a dynamic table that maps addresses to ports (interfaces) automatically.
- To make a table dynamic, we need a switch that gradually learns from the frame movements.
- To do this, the switch inspects both the destination and the source addresses.
- The destination address is used for the forwarding decision (table lookup); the source address is used for adding entries to the table and for updating purposes.

Figure 5.86 *Learning switch*

Gradual building of Table

Address	Port

a. Original

Address	Port
71:2B:13:45:61:41	1

b. After A sends a frame to D

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4

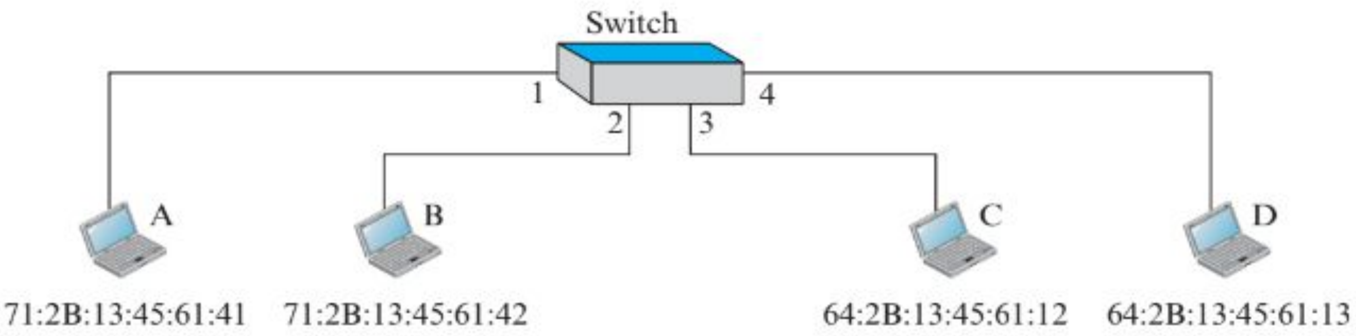
c. After D sends a frame to B

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4
71:2B:13:45:61:42	2

d. After B sends a frame to A

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3

e. After C sends a frame to D

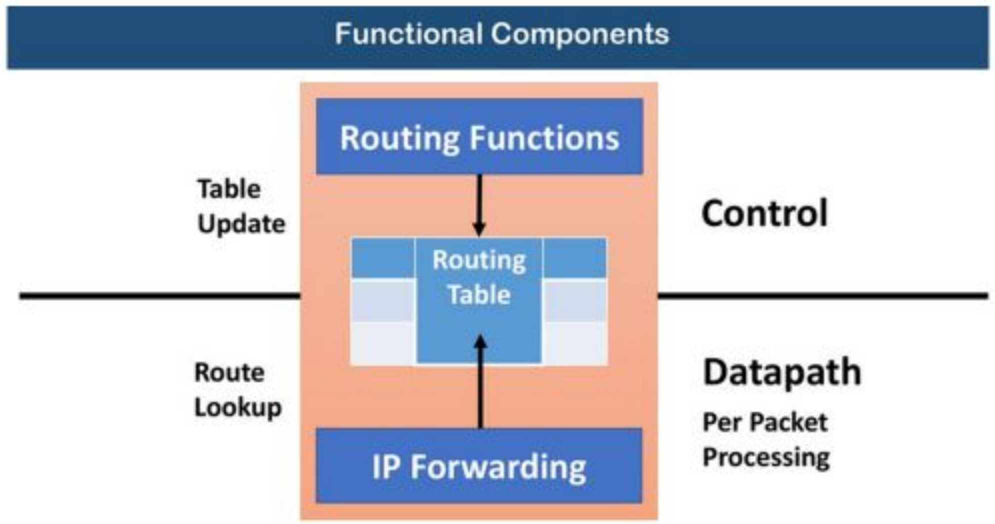


ROUTERS AND FIB

Router Hardware

- Processor is responsible for control functions **(route processors)**
  - Construct the routing table based on the routing algorithm
- Forwarding is done at the interface card
  - Route match needs to be very fast
  - Specialized hardware – Ternary Content-Addressable Memory (TCAM)

Control Plane	**Routing Table Construction**
Data Plane	**Forwarding**



Functional Components
Routing Functions
Table
Update
Control
Routing
Table
Route
Lookup
Datapath
Per Packet
Processing
IP Forwarding

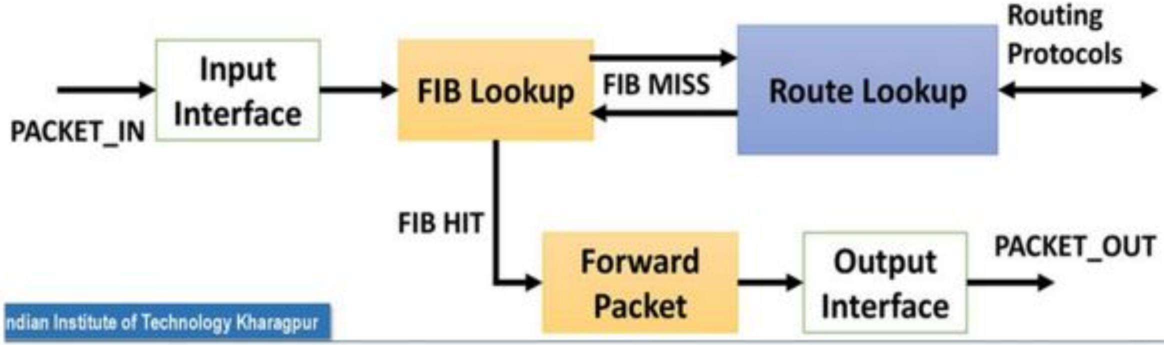
Routing Functions

- **Route Calculation**
- **Maintenance of the routing table**
- **Execution of the routing protocol**

- On commercial routers, routing functions are handled by a single general purpose processor, called the **route processor**

Forwarding Information Base (FIB)

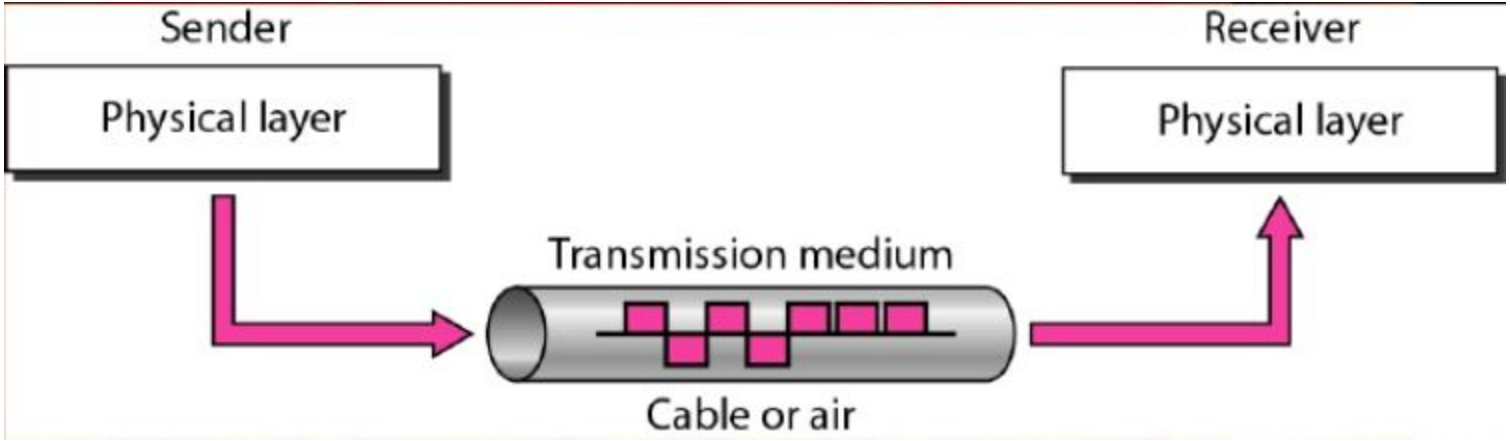
- The interfaces maintains a *forwarding information base* (FIB) – a mapping from input interface to output interface
- A replica of the routing table used at the interfaces for making the forwarding decision

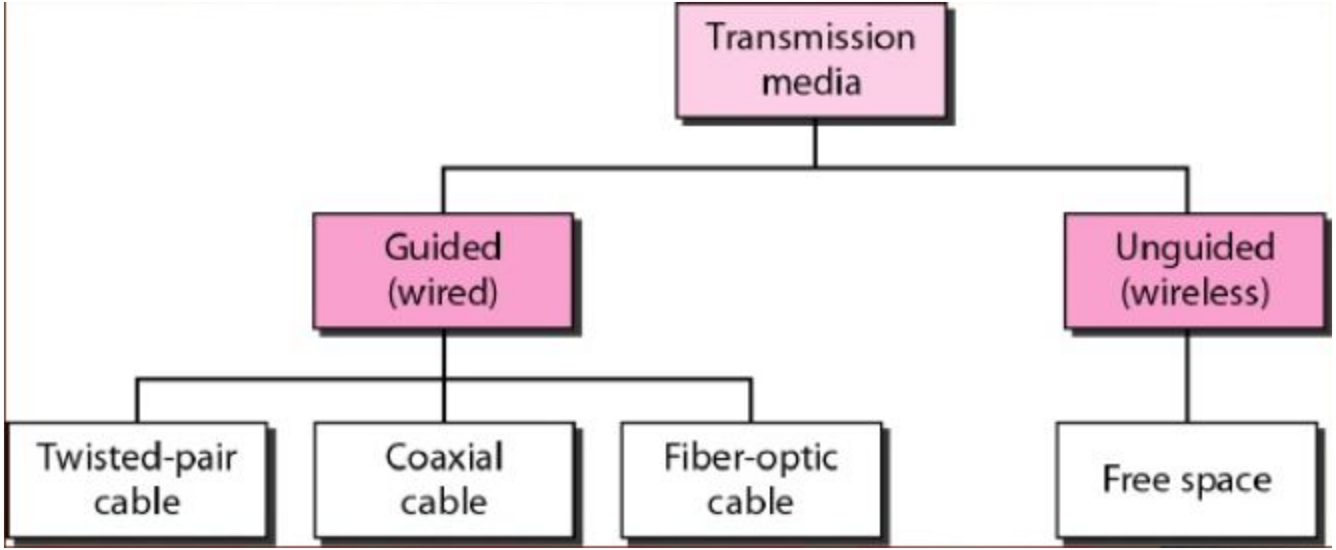


Difference between RIB and FIB

- **Routing Information Base (RIB)** – The routing table, implemented in software, is maintained at the control plane
- **Forwarding Information Base (FIB)** – The copy of the required routes maintained in interface TCAM hardware
- RIB is dynamic and maintains entire routing information, FIB is updated whenever required

TRANSMISSION MEDIA


Sender
Physical layer
Receiver
Physical layer
Transmission medium
Cable or air


Transmission media
Guided (wired)
Unguided (wireless)
Twisted-pair cable
Coaxial cable
Fiber-optic cable
Free space